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Lecture 4: Wasserstein, Duality

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Last Time

Review of last class

- Kantorovich formulation
- Couplings
- Birkhoff-vonn Neumann
- We showed that Brenier gives a deterministic coupling for the Kantorovich relaxation. We saw that Monge $\subset$ Kantorovich, and that Kantorovich is a convex relaxation of Monge.
- Define the feasible sets α and β
- **Interpretation:** average "work**?** to move mass $\mathbb{E}_\pi(c(X,Y))$
- **Existence:** We are guaranteed to have a solution, but it might not be unique

Today

- We are going to be talking about the Wasserstein distance, the metric properties of that distance, and duality (an alternate view of the problem)
- For a metric space (X,d), take $c(x,y) = d(x,y)^p, p \geq 1$. This yields the Wasserstein Distance:

$$W_p(\alpha,\beta) = \left(\inf_{\pi \in \prod(\alpha,\beta)} \int_X d(x,y)^p d\pi(x,y)\right)^{1/p}$$

For the Wasserstein distance, we take a distance and take its power to some p as defined above. The literature is mixed here, as some say all distances are Wasserstein distances.

Wasserstein distance was introduced in '69 in the context of functional analysis and Markov chains.

0.1 Example: Diracs

$\alpha = \delta_x, \beta = \delta_y$

- For the simple deltas, we just recover the ground metric (distance). Thus, it is *consistent* with the ground metric.

W_2 between 1D Gaussians

- $\alpha = N(\mu_\alpha, \sigma_\alpha^2), \beta = N(\mu_\beta, \sigma_\beta^2)$
- The closed form solution is just what we have seen in statistics classes – it looks like the Euclidean distance in the parameter space (the mean, variance space).

W_2 between ND Gaussians

- Closed form: $W_w(\alpha,\beta) = \|\mu_\alpha - \mu_\beta\|_2^2 + B(\Sigma_\alpha,\Sigma_\beta)^2$, where the second term is the *Bures* distance:

$$B(\Sigma_\alpha,\Sigma_\beta)^2 = \text{tr}\left(\Sigma_\alpha + \Sigma_\beta - 2(\Sigma_\alpha^{1/2}\Sigma_\beta\Sigma_\alpha^{1/2})^{1/2}\right)$$

Now, we have this object called the *Bures* distance, which is a function of the two covariance matrices and some matrix square roots, which looks like the Monge map we did a couple lectures ago.

If $\Sigma_\alpha, \Sigma_\beta$ commute, it simplifies to something akin to the Euclidean distance between the covariances:

$$B(\Sigma_\alpha, \Sigma_\beta)^2 = ||\Sigma_\alpha^{1/2} - \Sigma_\beta^{1/2}||_F$$

0.2 Gelbrich Bound on W_2

- Interpretation: For arbitrary measures α and β with means and covars, we can bound the Wasserstein distance using only the first two moments. For the Gaussian distributions, this is tight (an exact equality). Another way to interpret this is that Gaussian distributions are the distributions with the smallest distances for a given pair of first and second moments. This can be thought of the right side having the higher moments.
  - Note: the measure needs to have these moments in the first place.
- Applications
  - Useful as a cheap surrogate for W_2 in ML
  - Used in domain adaptation, distributional robustness, variational inference

W_2 factors out translations(constant shifts)

- Consider the translation operator $T_\tau : x \mapsto x - \tau$.

  This is most useful when you choose τ to be the mean of your distribution (i.e when you are centering your distribution), which means we can decompose their Wasserstein distance between the distance between their distributions and the distance between their means.

  $$W_2(T_{\tau\#}\alpha, T_{\tau'\#}\beta)^2 = W_2(\alpha, \beta)^2 - 2\langle \tau - \tau', \mu_\alpha - \mu_\beta \rangle + \|\tau - \tau'\|^2$$

  In particular, this implies

  $$W_2(\alpha, \beta)^2 = W_2(\tilde{\alpha}, \tilde{\beta})^2 + \|\mu_\alpha - \mu_\beta\|_2^2$$

  where $\tilde{\alpha}$ and $\tilde{\beta}$ are the centered versions of their respective measures. Some of these nicer, more simple results and simplifications are mostly for the 2 Wasserstein distance, and sometimes for the 1 Wasserstein distance. It's mostly L2 that gives us the nice thing.

Metric Properties

0.3 Metric Axioms

The important thing about Wasserstein distances is that they're actually distances in the mathematical sense, and they satisfy:

- **Non-negativity:** $W_p(\alpha, \beta) \geq 0$.
  - Proof: $d(x, y)^p \geq 0$ and $\pi(x, y) \geq 0$, inf of ≥ 0 numbers is ≥ 0.
- **Symmetry:** The couplings themselves are symmetric, which we require for metrics. $W_p(\alpha, \beta) = W_p(\beta, \alpha)$
  - Proof: $\pi \in \Pi(\alpha, \beta) \iff \pi^\top(A \times B) := \pi(B \times A) \in \Pi(\beta, \alpha)$. Also, $\int d(x, y)^p \, d\pi = \int d(x, y)^p \, d\pi^\top$ since $d(x, y) = d(y, x)$.
- **Identity:** Given $W_p(\alpha, \beta) = 0 \iff \alpha = \beta$.
  - Proof: One direction ($\Leftarrow$) is easy. For the other one: $\alpha \neq \beta$ implies any valid coupling has nonzero density in at least one pair $(x, y), x \neq y$, so infimum > 0.
- **Triangle inequality:** $W_p(\alpha, \beta) \leq W_p(\alpha, \eta) + W_p(\eta, \beta)$
  - Proof: Harder! Typical proof uses the gluing lemma. Given $\pi_{\alpha,\eta} \in \Pi(\alpha, \eta)$ and $\pi_{\eta,\beta} \in \Pi(\eta, \beta)$, define a *glued coupling* γ on $P(X, X, X)$ with 1-2 marginal $\pi_{\alpha,\eta}$ and 2-3 marginal $\pi_{\eta,\beta}$. This γ can be marginalized to give a coupling on α, β as well. Then use Minkowski and the triangle inequality of the ground metric.

So it's a real distance! So what?

- Mathematically:

  - This gives us a geometry on probability distributions, defining proximity etc.
  - Enables tools from functional analysis & geometry (gradient flows, PDE links)
  - W_p isn't just a number, it equips probability distributions with a metric space structure
- ML implications
  - Compare datasets in a stable way (unlike KL or TV)
  - **Guarantees continuity: small data changes → small change in distance**
  - Basis for algorithms: Wasserstein GAN, OT-based clustering, barycenters Once you have a distance, you can make statements about regularity and smoothness.
  - Thanks to these properties, it's been the basis for many algorithms in ML.

Wasserstein Space W_p

- We can endow $P(X)$ with the metric W_p
- Yields $\mathbb{W}_p := (P(X), W_p)$, the p-Wasserstein space (a metric space)
- This space is very nice – it's complete, separable, has existence of geodesics(important – the fact that there *is* a shortest path between two points) etc.
- A lot of nice theory already done for us!
- Why is this important?
  - Probability measures become points in a metric space, and it's nice because it allows us to view things in a "dataset" space.
  - Complete + Separable defines a Polish space (it's smooth without crazy irregularities etc.)
  - Provides a natural stage for ML tasks (datasets ↔ measures)

OT as a lifting operation

- Given a metric space (X, d) where data lives, and using OT with cost = d.
  - Lifting the point-wise metric d into a metric between distributions on that space
- The idea is we take a distance metric in the dataset space, and OT gives you a simple way to convert it to a distance between distributions defined in **that said space**. Thus, in order for OT to make sense, $d(x, y)$ needs to make sense – OT will not make sense with an irrelevant metric.
- OT geometry is "nice**?**

Weak convergence (weak* convergence and W_p)

- **Definition [weak* topology]:**
  We say that $(\alpha_k)_k$ converges weakly* to $\alpha \in P(X)$ (denoted as $\alpha_k \to \alpha$) iff for any continuous $f \in C(X)$,

$$\int_X f \, d\alpha_k \quad \to \quad \int_X f \, d\alpha.$$

- Theorem: Villiani 09, 6.9
  - $\alpha_k \rightharpoonup \beta \iff W_p(\alpha_k, \beta) \to 0$, aka W_p metrizes the weak* convergence
- **OT metrizes the weak* convergence,** i.e if the distance between two distributions converges to 0, then the metrics converge to each other.

Why geometry awareness matters

- TV drops suddenly, whereas W_2 is linear
  - KL undefined if supports don't overlap

- TV don't consider geometry
- OT/Wasserstein respects underlying space!

Dual Problem

Motivation for Duality

- The primal problem is what we have seen, which is minimization
- There's an alternative, complementary problem called the dual problem (maximization)
- They are two sides of the same coin.
- Every optimization problem can be associated with a dual, though not always useful, nor equivalent to primal

For LP

- We have a guarantee for equivalence, "strong duality"
- It gives an equivalent but more often than not more interpretable perspective
- Another route for algorithms!

Primal Problem		**Dual Problem**		
$\min_{\pi \in \mathbb{R}_+^{n\times m}}$	$\langle C, \pi \rangle$	$\max_{f\in\mathbb{R}^n, g\in\mathbb{R}^m}$	$a^\top f + b^\top g$	
s.t.	$\pi \mathbf{1} = a$	s.t.	$f_i + g_j \le c_{ij}$,	$\forall i, j$
	$\pi^\top \mathbf{1} = b$			

- The vectors f, g are called "dual potentials**?** (or Kantorovich, if optimal)
- Symmetries
  - We have switched up the locations of the constraints and objectives (fundamental aspect of duality)
  - We went from $n \times m$ variables to $n + m$ variables, but the constraints went the other way.

Cost vs Price interpretation

- Primal (transportation plan)
  - Sender wants to minimize shipping cost
  - Objective: min total shipping cost
  - Constraint: supply/demands are met
- Dual (potentials)
  - Vendor wants to maximize revenue (vendor says, outsource shipping to me)
  - Objective: maximize total revenue
  - Constraint: price never exceeds shipping cost (thus my vendor price must be competitive, otherwise I won't beat the market)

Primal Problem

$$\inf_{\pi\in\Pi(\alpha,\beta)} \int_{X\times Y} c(x,y)\, d\pi(x,y) \qquad \text{where}$$

$$\Pi(\alpha,\beta) = \left\{\pi \in \mathscr{P}(X\times Y) \;\middle|\; P_{x\#}\pi = \alpha,\ P_{y\#}\pi = \beta\right\}$$

Dual Problem

$$\sup_{(f,g)\in R(c)} \int_X f(x)\, d\alpha(x) + \int_Y g(y)\, d\beta(y) \qquad \text{where}$$

$$R(c) = \{(f, g) \in C(X) \times C(Y) \mid \forall (x, y),\ f(x) + g(y) \leq c(x, y)\}$$

Two special cases

- 1-Wasserstein distance ($d(x, y)$)
  - In that case, you lose the g, and it becomes $-f$. They collapse to a single 1-Lipschitz test function f.
  - Basis of Wasserstein GAN critic
  - The potential is a clear intuition of "adversary".
- 2-Wasserstein distance
  - Becomes **convex**. Related to Brenier, OT becomes gradient of a convex function.

Corollary of Brenier: The optimal dual potentials and φ are related by:

$$\varphi(x) = \tfrac{1}{2}\|x\|^2 - f(x)$$

$$\psi(x) = \tfrac{1}{2}\|x\|^2 - g(x)$$

where ψ is cvx and $T^{-1} = \nabla\psi$.

Applications in ML

- Generative Modeling (Wasserstein GANs) Stable training, more meaningful gradients than other GANs.
- Neural OT: stable learning of Monge maps via dual potentials Most of the methods for learning Monge/OT maps use the dual form of the task.
  - Example applications: style transfer, domain adaptation
- Other uses:
  - Distributionally robust optimization (DRO): worst case loss over Wasserstein ball